Genome as Projection: Coupling Channels Predict Cancer Survival and Suggest Drug Response Principles

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Impact statement: Counting disrupted organizational channels, not mutated genes, predicts cancer survival across 73,593 patients and identifies untested drug targets in 4,000 additional patients.

Abstract

The genome is a linear projection of a functional graph. Genes that appear distinct in sequence can be copies of the same functional node, forced into separate genomic addresses by serialization (projection families). The distributed connectivity between them, too diffuse to encode in any single gene, constitutes coupling channels.

Channel count predicts overall survival across 73,593 patients; mutation count does not. Genes in the same projection family carry the same drug vulnerability: expanded PARP, PI3K/AKT, and FGFR eligibility, combination timing principles, and explanations for published trial failures follow directly. Channel severance by any mechanism produces predictable cancer risk: transplant immunosuppression, neurodegeneration, information-environment exposure.

Synthetic lethality mutual exclusivity varies fivefold across cancer types, mapping directly to tissues where PARP inhibitors show clinical efficacy. The co-mutation matrix alone identifies which channels are load-bearing in which tissues.

Introduction

The functional relationships between genes form a dense, cyclic graph. The genome is a linear sequence. Serializing one into the other forces copies: a single genomic address cannot serve all adjacencies of a hub node. Paralogs, gene families, "functionally related" genes are projection artifacts. They are copies of the same node at different addresses.

Three claims, decreasing certainty.

Demonstrated. Channel count predicts survival across 73,593 patients ($p < 10^{-44}$). Mutation count does not. More mutations in fewer channels means deeper commitment to one escape route: a more constrained, treatable tumor.

Interpretive. A coupling channel is not a pathway. It is the organizational function a pathway serves. Channels are the connectivity too distributed to precipitate into any single gene.

Predictive. Genes in the same projection family carry the same vulnerability. A drug that works on one works on all. Channels severed by any mechanism produce the same cancer risk.

Channels connect cells to tissue, organ, and organism levels. Severing them forces cell level optimization. Growth control falls first; immune surveillance last. Formal derivations: [5, 6, 7, 8]. This paper can be read independently.

Results

Channel count predicts survival; mutation count does not

Three publicly available datasets from cBioPortal [15] were analyzed: the MSK-IMPACT Clinical Sequencing Cohort (n = 7,091) [16], the MSK MetTropism Cohort (n = 24,646) [17], and the MSK-IMPACT 50K Cohort (n = 41,856) [18]. Together: 73,593 patients, 28,750 deaths.

In multivariate Cox models, channel count is significant in every dataset ($p < 0.005$). Mutation count either drops out (MSK-IMPACT 2017: $p = 0.67$) or becomes *protective* (MetTropism: HR = 0.82; 50K: HR = 0.83). If channel count proxied for tumor burden, mutation count would predict worse outcomes. It predicts better (Table 1, Figure 1).

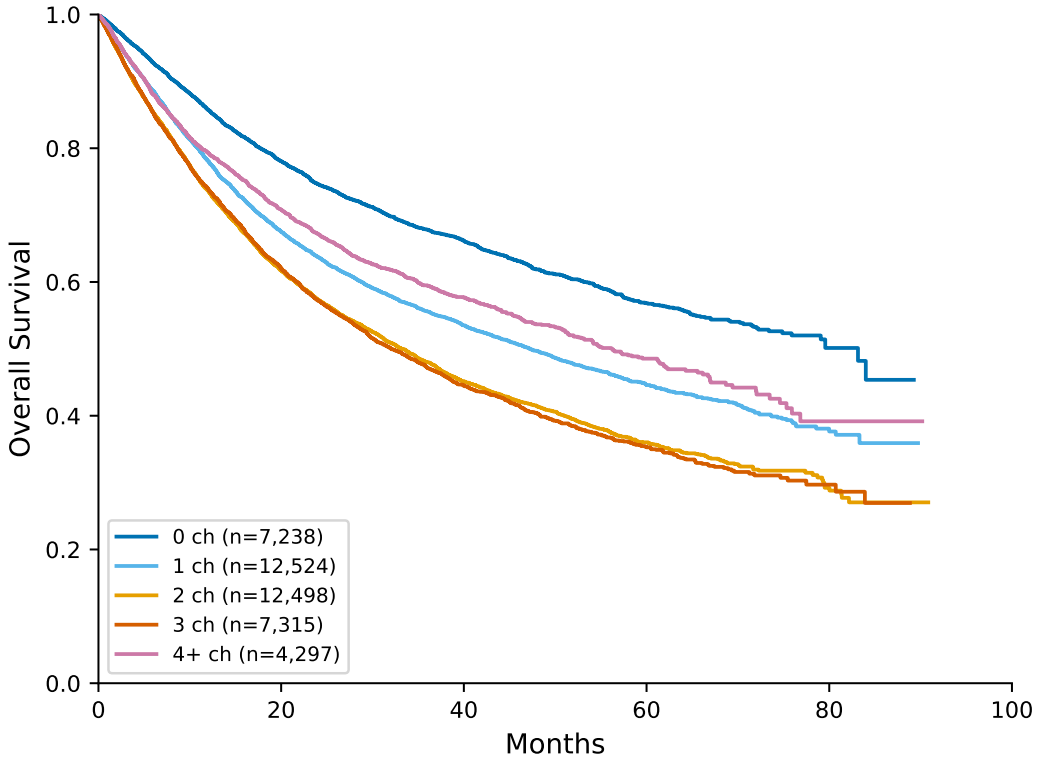

Figure 1: Kaplan–Meier survival curves stratified by the number of coupling channels severed (MSK-IMPACT 50K, n = 43,872). Survival decreases monotonically from 0 to 3 channels. The 4+ group shows partial rebound, consistent with hypermutation phenotypes where most mutations are passengers.

Kitchen-sink Cox model with 275 covariates (TMB, MSI, cancer type, subtype, age, sex, tumor purity, FGA, primary site, sample type, panel version): channel count HR = 1.09, $p = 10^{-23}$. MetTropism (187 covariates): HR = 1.11, $p = 10^{-18}$. MSK-IMPACT 2017 (109 covariates): HR = 1.14, $p = 2 \times 10^{-8}$.

Table 1: Cox proportional hazards: channel count vs. mutation count across three independent cohorts totaling 73,593 patients. HR = hazard ratio per unit increase. In multivariate models, channel count remains significant while mutation count loses predictive power or becomes protective.

Dataset	Predictor	HR	p	Note
MSK-IMPACT 2017	Channel count (univariate)	1.10	<0.005	
(n = 7,091)	Log mutation count (univariate)	1.12	<0.005	
	Channel count (multivariate)	1.09	<0.005	Significant
	Log mutation count (multivariate)	1.02	0.67	Not significant
MetTropism	Channel count (univariate)	1.06	<0.005	
(n = 24,646)	Log mutation count (univariate)	1.00	0.69	Not significant
	Channel count (multivariate)	1.17	<0.005	Significant
	Log mutation count (multivariate)	0.82	<0.005	Protective
MSK-IMPACT 50K	Channel count (univariate)	1.04	<0.005	
(n = 41,856)	Log mutation count (univariate)	0.98	0.03	Weakly protective
	Channel count (multivariate)	1.15	<0.005	Significant
	Log mutation count (multivariate)	0.83	<0.005	Protective

Cross-channel co-mutations are worse than same-channel

Two mutations in different channels produce worse outcomes than two mutations in the same channel (Table 2). The multivariate Cox model controls for mutation count: channel count is significant after adjustment.

Table 2: Cross-channel vs. same-channel co-mutation mortality. Cross-channel: 2+ mutations hitting different coupling channels. Same-channel: 2+ mutations confined to one channel.

Dataset	Cross-channel	Same-channel	p (log-rank)
MSK-IMPACT 2017 (n = 7,091)	33.1%	27.7%	1.4×10^{-5}
MetTropism (n = 24,646)	42.9%	39.0%	6.3×10^{-13}
MSK-IMPACT 50K (n = 41,856)	42.6%	39.7%	5.7×10^{-13}

Stratifying by fixed channel count, patients with more mutations survive longer. At 2 channels: mortality drops from 46.2% to 41.2% ($p = 3.1 \times 10^{-8}$). At 3 channels: 46.9% to 38.6% ($p = 7.2 \times 10^{-14}$). An alternative explanation is immunogenicity. If neoantigen load explained the protective effect, it should also protect cross-channel patients. It does not (Table 2). The protection is specific to same-channel depth, not total mutation burden.

Synthetic lethality is tissue-conditional

Cells that lose both copies of a functional node die. SL gene pairs should therefore be mutually exclusive in tumors, but only where the channel is load-bearing.

Figure 2: 121 SL pairs, 41 cancer types (n = 44,283). Fivefold variation. Ovarian (obs/exp = 0.74), NSCLC (0.76), squamous lung (0.87): strong exclusivity. These are where PARP inhibitors work. Colorectal (2.60), endometrial (2.97): strong co-occurrence. Both MSI-high: mismatch repair

already gone, DDR array already failed, exclusivity constraint relaxed. The co-mutation matrix alone maps which channels are load-bearing in which tissues.

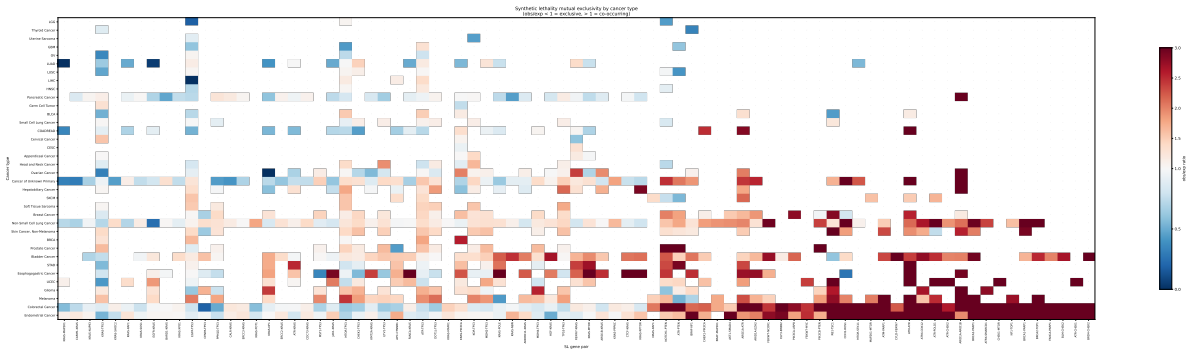

Figure 2: Synthetic lethality mutual exclusivity by cancer type. Each cell shows the observed/expected co-mutation ratio for an SL gene pair in a given cancer type. Blue (obs/exp < 1): mutually exclusive, the channel is load-bearing. Red (obs/exp > 1): co-occurring, the channel is bypassed or the tumor is hypermutated. Cancer types are sorted by mean exclusivity. The pattern predicts where PARP inhibitors should be effective (blue rows) and where they should not (red rows).

Channels are tissue-specific

Cancer type as covariate (38 types) attenuates channel count by 1.1% (HR $= 1.17$, $p = 8 \times 10^{-73}$). Within-type: NSCLC (HR $= 1.16$), breast (1.29), prostate (1.47), pancreatic (1.32), all $p < 10^{-10}$. Not a proxy for cancer type. Figure 3: breast is PI3K/Growth and Endocrine. Glioma is Cell Cycle and DDR. Colorectal is Tissue Architecture and PI3K/Growth.

Severance hierarchy: cell-intrinsic channels first

Not random (Figures 4, 5). One channel severed ($n = 12{,}558$): 80% PI3K/Growth or Cell Cycle. Endocrine and Immune: under 5%.

PI3K/Growth $\rightarrow$ Cell Cycle $\rightarrow$ Tissue Architecture $\rightarrow$ DDR $\rightarrow$ Endocrine $\rightarrow$ Immune. Growth escape is the gateway; immune evasion is last. The co-severance matrix (Figure 5) is asymmetric: cell-intrinsic severance predicts organism-level, not the reverse.

1.9% of multi-channel patients lack a cell-intrinsic gateway. Enriched 6.5$\times$ for prostate, 2.5$\times$ breast, 7.6$\times$ mesothelioma: cancers entering through organism level channels. Lower mortality (34.8% vs. 42.6%). They never escaped the most fundamental coupling layer.

Sensitivity and robustness

The signal peaks at 3 channels ($p = 3 \times 10^{-124}$) and is significant from 2 through 8 (Table 3). At 10, over-splitting collapses channel count back into a proxy for mutation count. Hierarchical clustering on Jaccard co-mutation distances produces groupings that fail: at $k = 6$, HR $= 0.80$ ($p = 9 \times 10^{-6}$), the wrong direction. Biological channels at $k = 6$: HR $= 1.10$ ($p = 3 \times 10^{-9}$). ARI between data-driven and biological groupings: 0.018. Co-mutation clusters group genes by which tumors they appear in. Channels group genes by which organizational system they serve.

Table 3: Sensitivity analysis (MSK-IMPACT 50K, $n = 41{,}856$): multivariate Cox hazard ratios at different channel granularities.

Channels	Ch HR	Ch p	Mut HR	Mut p
2	1.25	2×10^{-52}	0.94	4×10^{-9}
3	1.30	3×10^{-124}	0.86	8×10^{-41}
4	1.21	3×10^{-75}	0.88	9×10^{-27}
5	1.17	8×10^{-58}	0.88	6×10^{-22}
6	1.14	1×10^{-44}	0.90	2×10^{-16}
7	1.10	7×10^{-24}	0.93	1×10^{-7}
8	1.07	3×10^{-14}	0.95	4×10^{-4}
10	1.01	0.064	1.02	0.20

Restricting to truncating mutations strengthens the signal: HR rises from 1.15 to 1.19 ($p = 10^{-55}$). MSK-IMPACT sequences 400–500 genes, systematically undercounting channel hits. This works against the hypothesis, not for it.

Predictions from the projection framework

DDR family: PARP eligibility expansion. Current trials screen for 9 DDR genes. Ten additional genes share the same functional role: *BLM*, *NBN*, *ERCC4*, *CHEK1*, *CHEK2*, *RAD51B*, *RAD51*, *BRIP1*, *MRE11*, *XRCC2*. Over 4,000 patients in the MSK-IMPACT cohort. None currently eligible for PARP trials.

PI3K and FGF family expansion. *INPP4B*, *PIK3CB*, *PIK3CD*: AKT inhibitor sensitivity comparable to *PTEN*-loss. *FGFR4*: pan-FGFR inhibitor response. *CREBBP* and *EP300* (3,491 combined patients): functionally equivalent, same future targeted therapy.

Same-channel combinations fail. PARP + ATR, PARP + WEE1, PARP + ATM, dual PARP1/2 versus PARP1-selective: no significant PFS improvement over the simpler regimen. Two drugs in one channel add dose, not a new therapeutic axis.

Cross-channel timing matters. Metastatic HR+ breast cancer post-endocrine progression: adding a SERD to a PARP inhibitor buys little. The endocrine channel is already gone. Adjuvant setting, different story: low-dose PARP1-selective alongside endocrine therapy in germline DDR carriers. The PARP kills cells at DDR second-hit before they evolve ESR1 resistance. The endocrine therapy keeps the channel open. Each drug extends the other. PARP1-selective agents eliminate PARP2-driven hematological toxicity, enabling long-term adjuvant dosing.

Natural PARP1 disruption. *PARP1* loss-of-function alongside DDR partner mutations: the genome produced the same synthetic lethality the drug exploits. These patients should show improved survival without PARP inhibitor treatment.

Graph position determines phenotype. Hub mutations carry worse prognosis than leaf mutations in the same channel. BRCA1 (DDR hub) drives triple-negative breast cancer. BRCA2 (DDR leaf) drives HR+. RAD51C phenocopies BRCA2, not BRCA1. GOF and LOF in the same gene produce opposite signatures.

Severance is directional at two scales. Across channels: growth first, immune last (Figure 5). Within channels: leaves before hubs (*CDK4* before *TP53*, *BRCA2* before *BRCA1*, *BRAF* before *KRAS*). Reverse direction at either scale: rare, better prognosis.

Non-genetic severance. Transplant immunosuppression severs the immune channel artificially. Cancer risk elevates in immune-dependent types, not uniformly [11]. Parkinson's: reduced cancer risk (context-switching machinery disabled, fewer DNA damage events). ADHD: elevated cancer risk (excessive switching, chromatin damage accumulates). Stimulants should reduce it. Early-onset cancers in young adults: enriched for chromatin remodeling mutations.

Companion papers: [9] (graph position), [10] (non-genetic severance).

Discussion

Genome as projection

Serialization forces three compromises. Cycles must be broken: feedback loops are severed during projection and reconstructed at runtime via transcription factors, signaling cascades, chromatin looping. These reconstructed edges are the most fragile. Hub nodes must be copied: a single address cannot serve all adjacencies. What appears as genetic diversity is serialization overhead. Distributed connectivity cannot be written at all. Channels exist only in the three dimensional fold of the chromatin: unprecipitated knowledge, reconstructed by physical proximity when the helix folds.

Chain severance

A cell maintains finite regulatory states (C_n) [2]. Multicellularity overcomes this by hierarchical coupling: cell to tissue to organ to organism [6]. Channels are the dimensions: gap junctions [1], paracrine signaling, endocrine and immune [4].

Lose a channel and the cell's context contracts. What remains: proliferation, metabolic autonomy, motility. Hanahan and Weinberg's hallmarks [4] are not acquired capabilities. They are what a cell does when it can no longer hear the tissue. The Warburg effect is not a metabolic choice; it is what happens when oxygen supply information is gone [14].

Projection families

The genome maintains redundant copies of critical subgraphs like a RAID array. Lose one copy and the system degrades. Remove the parity check (the synthetic lethal partner) and the degraded array fails.

A PARP inhibitor works regardless of which DDR copy is broken: BRCA1, BRCA2, RAD51C. The drug targets function, not address. Severance occurs through multiple mechanisms: genetic, microenvironmental, epigenetic, spatial (Table 4). All produce the same endpoint. This is why hereditary, environmental, and sporadic cancers converge.

Prior work

TOFT [12] got the level right (tissue, not cell) and the default state right (proliferative). It lacked formalism. The escalation chain provides it. No existing cancer model addresses projection. Current frameworks treat genes as independent entities in shared pathways. They are copies of shared nodes. Different question, different answer to "target a gene."

Limitations

Stage at diagnosis and treatment history are not available in these datasets. Channel count could correlate with stage. Three partial controls mitigate this: (1) within-cancer-type analysis, where stage distributions are more homogeneous; (2) the 275-covariate model includes FGA and tumor purity, which are stage-correlated proxies; (3) MetTropism is an all-metastatic cohort where stage variation is minimal, and the signal replicates. None fully substitute for stage adjustment. The projection family predictions are untested. The non-genetic severance predictions are derived from the framework but not validated in this study.

Materials and methods

Datasets

Three publicly available datasets were obtained from cBioPortal [15]: the MSK-IMPACT Clinical Sequencing Cohort (n = 7,091) [16], the MSK MetTropism Cohort (n = 24,646) [17], and the MSK-IMPACT 50K Cohort (n = 41,856) [18].

Channel mapping

Somatic mutations were mapped to six coupling channels based on the organizational function of the mutated gene: (1) DNA damage response (DDR), (2) cell cycle checkpoint, (3) growth signaling (PI3K–AKT–mTOR, RAS–MAPK), (4) endocrine/hormone, (5) immune surveillance, (6) tissue architecture. Non-silent mutations (missense, nonsense, frameshift, splice site) were included. For each patient, the number of distinct channels with at least one mutation was recorded. The complete gene-to-channel mapping (92 genes across 6 channels) is provided in Table 6 and the supplementary repository.

Statistical analysis

Cox proportional hazards models were fitted with channel count and log mutation count as predictors. Multivariate models included both predictors simultaneously. Cancer-type adjustment used one-hot encoding of 38 cancer types. The kitchen-sink model included all available covariates (TMB, MSI, cancer type, subtype, age, sex, tumor purity, FGA, primary site, sample type, panel version).

Sensitivity analysis repeated the multivariate Cox regression at channel granularities from 2 to 10. Data-driven channel groupings were produced by hierarchical clustering on Jaccard co-mutation distances and compared to biological groupings using the Adjusted Rand Index.

Synthetic lethality analysis

Synthetic lethal gene pairs were obtained from BioGRID and SynLethDB. For each SL pair and cancer type with $\geq$ 200 patients, the observed/expected co-mutation ratio was computed as obs/exp $= n_{\text{both}}/(f_A \times f_B \times N)$ where f_A and f_B are the per-gene mutation frequencies and N is the number of patients. Pairs with expected co-occurrence < 2 were excluded.

Code and data availability

All analysis code is available at `https://github.com/patdmccarthy/open-knowledge-graph`. All datasets are publicly available through cBioPortal. The knowledge graph (Neo4j) and all intermediate results are archived on Zenodo.

Acknowledgments

The author used Claude (Anthropic) for drafting assistance, literature review, and LaTeX preparation. All theoretical content, proofs, and arguments are the author's own.

Competing interests

The author declares no competing interests. The author's spouse is a participant in the PETRA clinical trial (AstraZeneca), which is discussed in the predictions section.

Author contributions

Patrick D. McCarthy: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization.

Funding

This research received no external funding.

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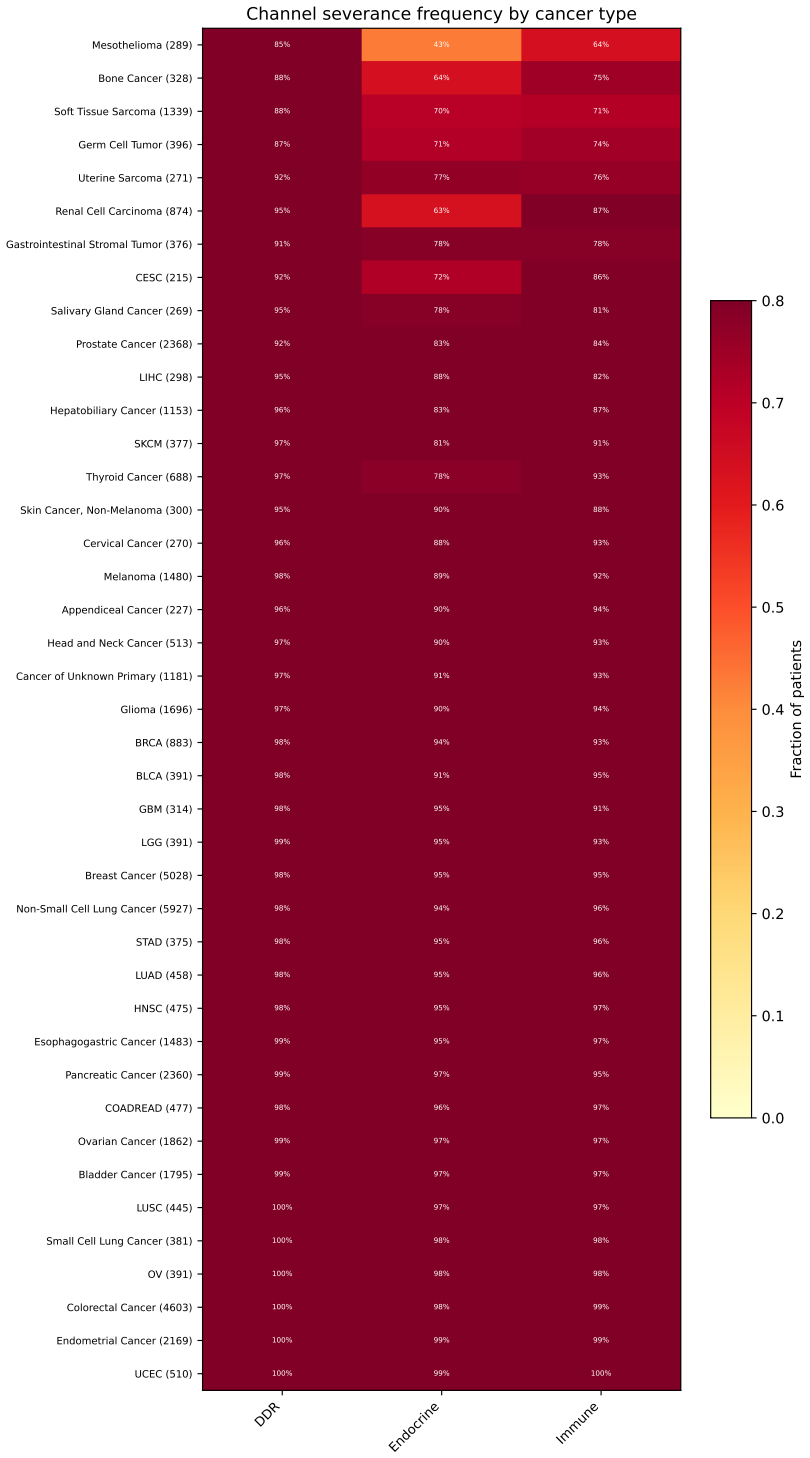

Figure 3: Channel severance frequency by cancer type. Each cell shows the fraction of patients with at least one mutation in the given channel. Cancer types sorted by total channel burden. Breast is Endocrine-dominant, colorectal is Tissue Architecture-dominant, glioma is Cell Cycle-dominant.

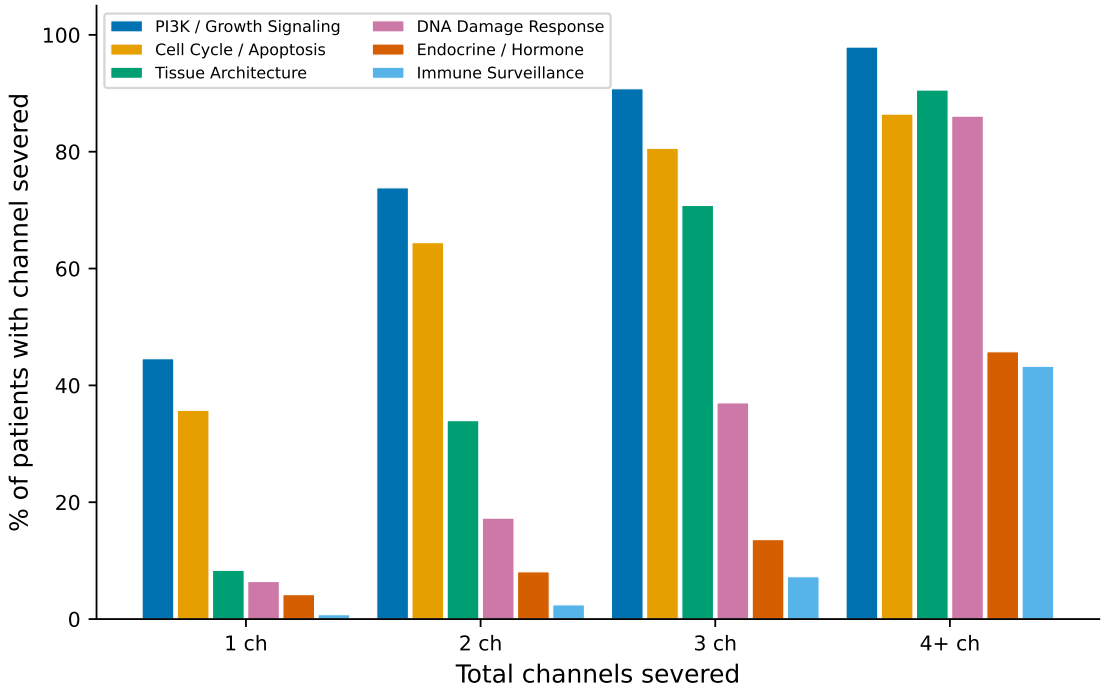

Figure 4: Severance hierarchy across channel-count strata (MSK-IMPACT 50K). Each bar shows the percentage of patients at a given stratum who have that channel severed. Cell-intrinsic channels dominate at every stratum; organism-level channels appear only at higher channel counts.

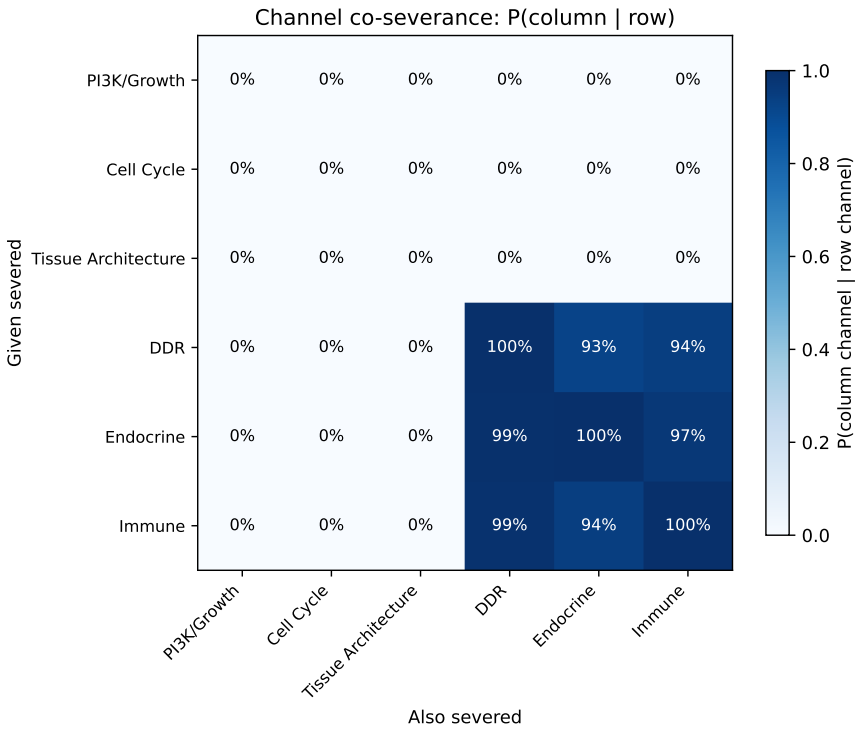

Figure 5: Channel co-severance matrix. Each cell shows P(column channel severed | row channel severed). The matrix is asymmetric: cell-intrinsic channels predict organism-level severance, but not the reverse. Severance propagates inward toward the root of the escalation hierarchy.

Tables

Table 4: Mechanisms of chain severance. All four produce the same outcome: loss of coupling, context collapse, and cell-level optimization.

Mechanism	How coupling is lost	Example	Identified by
Genetic	Cell's receivers damaged	E-cadherin loss, p53/Rb mutation	SMT [13]
Microenvironmental	Signal no longer transmitted	Chronic inflammation, fibrosis, tissue damage	TOFT [12]
Epigenetic	Receiver genes silenced without sequence change	Methylation of connexin or cadherin promoters	—
Spatial	Cell leaves signaling field	Basement membrane breach, metastasis	—

Table 5: Relationship to existing cancer frameworks.

Framework	What it captures	Reinterpretation
Hallmarks [4]	Universal capabilities across cancers	Not acquired but *revealed*: necessary action space under cell-level bounded context
Atavistic theory [3]	Phylostratigraphic gene pattern	Convergent necessity, not historical reversion
Clonal evolution [19]	Darwinian selection dynamics	Provides the *attractor*: direction determined by bounded context, not random
Attractor landscape [20]	Cancer as abnormal attractors	"Abnormal" attractors are the *default* under cell-level bounded context
Cancer as cheating [21]	Cooperation breakdown	Cheating is not a strategy but the necessary outcome of chain severance
Ecological process [22]	Tumor as ecosystem	Correct dynamics, but missing why they converge

Table 6: Complete gene-to-channel mapping used in all analyses (92 genes across 6 channels). Genes are assigned based on organizational function served. The full mapping and analysis code are provided in the supplementary repository.

Channel	**Genes**
DNA Damage Response	*ATM*, *ATR*, *BRCA1*, *BRCA2*, *PALB2*, *RAD51C*, *RAD51D*, *RAD51B*, *CHEK1*, *CHEK2*, *BAP1*, *BARD1*, *FANCA*, *FANCC*, *FANCD2*, *MLH1*, *MSH2*, *MSH6*, *PMS2*, *POLE*, *POLD1*
Cell Cycle / Apoptosis	*TP53*, *RB1*, *CDKN1A*, *CDKN1B*, *CDKN2A*, *CDKN2B*, *CDK4*, *CDK6*, *CCND1*, *CCNE1*, *MDM2*, *MDM4*, *MYC*, *MYCN*
PI3K / Growth Signaling	*PIK3CA*, *PIK3R1*, *PTEN*, *AKT1*, *AKT2*, *AKT3*, *MTOR*, *KRAS*, *NRAS*, *HRAS*, *BRAF*, *RAF1*, *MAP2K1*, *MAP2K2*, *MAP3K1*, *MAP3K13*, *ERBB2*, *ERBB3*, *EGFR*, *FGFR1*, *FGFR2*, *FGFR3*, *IGF1R*, *MET*, *NF1*, *NF2*, *TSC1*, *TSC2*, *STK11*, *ARID1A*
Endocrine / Hormone	*ESR1*, *ESR2*, *PGR*, *AR*, *FOXA1*, *GATA3*
Immune Surveillance	*B2M*, *HLA-A*, *HLA-B*, *HLA-C*, *JAK1*, *JAK2*, *STAT1*, *CD274*, *PDCD1LG2*, *CTLA4*
Tissue Architecture	*CDH1*, *CDH2*, *CTNNB1*, *APC*, *AXIN1*, *AXIN2*, *SMAD2*, *SMAD3*, *SMAD4*, *TGFBR1*, *TGFBR2*, *NOTCH1*, *NOTCH2*, *NOTCH3*, *NOTCH4*, *FBXW7*, *GJA1*, *GJB2*